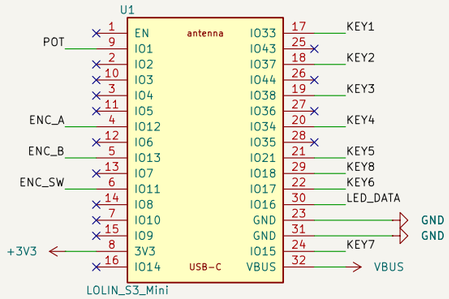
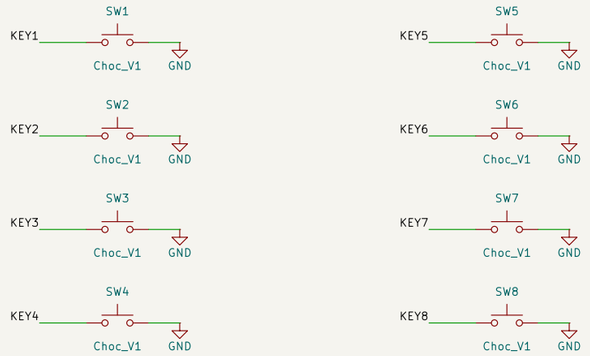
MCU — LOLIN S3 Mini

Module sits on the BACK of the PCB (USB-C at the chin edge).
Unused pins are no-connect. IO3 = strapping, IO43/44 = UART console, IO35/36 kept free for I2C.



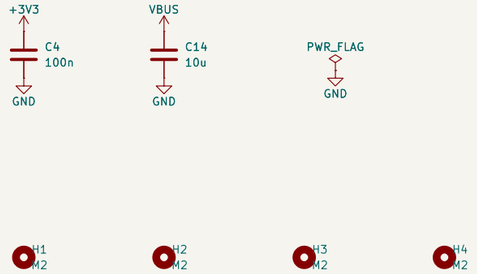
Keys — 8x Kailh Choc V1

Direct to GPIO, active low, internal pull-ups in firmware.
No diodes needed (no matrix). Face position is chosen at placement.



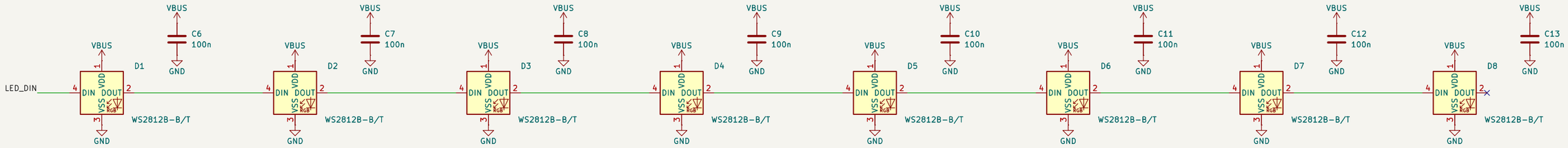
Power + mounting

VBUS = USB 5 V straight from the module (no fuse).
C4 next to U1 3V3 pin, C14 bulk at the LED chain entry.



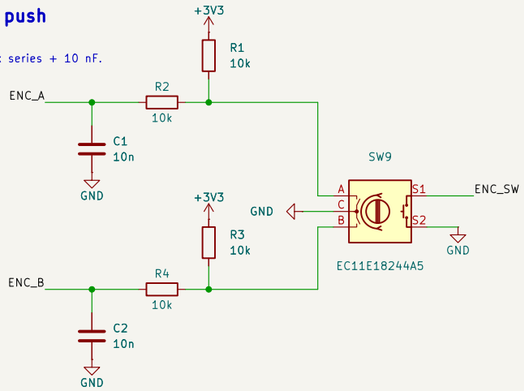
Underglow — 8x WS2812B-B/T (bottom side)

100 nF per LED. Data enters at D1 from the level shifter (LED_DIN). Cap brightness in firmware (~36 mA/LED full white).



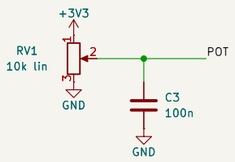
Rotary encoder — EC11 with push

EC11E18244A5 (36 detents / 18 pulses).
Bourns filter per channel: 10k pull-up + 10k series + 10 nF.



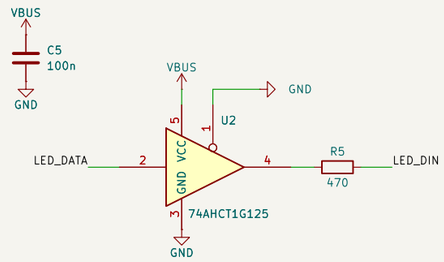
Pot — 10k linear

PTV09A-4020F-B103.
Wiper -> IO1 (ADC1).



LED data level shift — 3.3 V to 5 V

74AHCT1G125: TTL-level input (VIH 2.0 V) at 5 V.
Generic WS2812B need 0.7*VDD = 3.5 V.



Phase 2 schematic. Parts + LCSC numbers: docs/parts.md
LOLIN S3 Mini (ESP32-S3) — 8 Kailh Choc V1 — EC11 — 10k pot — 8x WS2812B underglow
Hack Club Half Life

Sheet: /
File: knob.kicad_sch

Title: Knobpad — panda MIDI keypad

Size: A3	Date: 2026-10-24	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/1